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Glue gun

Abstract

A glue gun, including: a main body with one end formed as an accommodation part for accommodating a glue cylinder; a fixed handle connected to another end of the main body; a movable handle pivotally connected to the fixed handle; a push rod with one end located in the accommodation part; a brake provided on the push rod; and a movable member provided on the main body, where the movable member is configured to lock the brake, so as to continuously apply a force on the glue cylinder, and release the brake to release the force applied by the brake on the glue cylinder.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to the field of hardware, and in particular to a glue gun, which belongs to the technical field of tools for use in production, processing, manufacturing and construction industries.

DESCRIPTION OF THE PRIOR ART

(2) During the use of a glue gun, glue is pushed by a gun-type handle lever, which continuously reciprocates to squeeze the glue such that same is discharged. One end of the glue gun is provided with a trigger mechanism suitable for a user to hold and press. As the user manually presses the trigger mechanism, the glue placed in the glue gun can be driven by a mechanical structure of the glue gun such that the glue is discharged at a glue outlet of a glue cylinder. One glue gun can be adapted to various types of glue with different characteristics, and different glue has different fluidity. A common glue gun only has a fixed squeezing mechanism, so it is very difficult to achieve uniform glue discharge in the process of glue application. In specific application scenarios of the glue gun, different glue discharge speeds are required owing to different ranges of glue application. However, the common glue gun only has the fixed squeezing mechanism, so the glue discharge speed of the glue cylinder can be controlled only by means of the pressing speed of the user's hand. In fact, it is extremely difficult to control the pressing speed of the user's hand while applying a force, especially when applying a large force to the handle. In the process of starting to use the glue gun, usually, since same is left unused for a long time, some of the glue is solidified, and the static viscosity of the glue and other problems exist, a relatively large force is usually required to drive the glue gun in the beginning of using same, and after the glue discharge is stabilized, the driving force required by the glue gun will be significantly reduced. The fixed squeezing mechanism of the common glue gun cannot solve these problems. In addition, in order to discharge the glue, same is usually squeezed hard in the beginning of using the glue gun, but the squeezing speed is difficult to control, so there is the problem of glue waste caused by excessive extrusion after the glue is squeezed out.

(3) Therefore, a person skilled in the art is dedicated to developing a glue gun, in which the gear can be adjusted to enable switching between a fast feed and a powerful feed, thereby enabling the user to make adjustment according to actual requirements.

SUMMARY OF THE INVENTION

(4) In view of the above defects in the prior art, the technical problem to be solved by the present invention is how to make gears of the glue gun adjustable.

(5) In order to achieve the above-mentioned purpose, the present invention provides a glue gun, where the glue gun includes a movable handle and a fixed handle, and the movable handle is connected to fixed handle via a pivot pin shaft; the movable handle is further provided with an actuating member, and the actuating member is provided with an actuating part cooperating with a push member; and a spacing between the pivot pin shaft and the actuating part is sized to be switchable between two or more gears, such that the magnitude of the force applied by the actuating member on the push member is changed.

(6) Further, the movable handle is provided with a sliding groove, which is provided with two or more gears; and the pivot pin shaft is arranged to be slidable in the sliding groove so as to enable switching between the gears.

(7) Further, the fixed handle is provided with a sliding groove, which is provided with two or more gears; and the pivot pin shaft is arranged to be slidable in the sliding groove so as to enable

switching between the gears.
(8) Further, an elastic component is further provided between the movable handle and the fixed handle; and the elastic component is arranged to produce a pre-tightening force on the movable handle so as to keep the pivot pin shaft in one of the gears.
(9) Further, the pivot pin shaft is a stepped pin shaft having a pressing end, the pressing end is sheathed with an elastic component, the stepped pin shaft has a first shaft diameter portion and a second shaft diameter portion, the shaft diameter of the first shaft diameter portion is greater than the width of the sliding groove, the shaft diameter of the second shaft diameter portion is smaller than the width of the sliding groove, and the stepped pin shaft is arranged such that: under the biasing action of the elastic component, the first shaft diameter portion is in the gear; when the pressing end is pressed, the stepped pin shaft moves axially such that the second shaft diameter portion goes into the gear; and when the pressing end is released, the stepped pin shaft moves axially in the opposite direction under the action of a restoring force of the elastic component such that the first shaft diameter portion goes into the gear again.
(10) Further, the sliding groove is an arc-shaped groove or a linear groove.
(11) Further, the movable handle is provided with a sliding groove, which is provided with two or more gears; and the actuating part is arranged to be slidable in the sliding groove so as to enable switching between the gears.
(12) Further, the actuating member further includes a pivot, and the actuating part of the actuating member is arranged to be slidable in the sliding groove around the pivot.
(13) Further, a brake is sheathed on a push rod, one end of the brake cooperates with a limiting groove of a main body, and the one end of the brake moves between a first limiting end and a second limiting end of the limiting groove, so that the push rod has an idle stroke, with a distance from the first limiting end to the second limiting end, in the pushing process.
(14) Further, the idle stroke is 3 to 5 mm.
(15) In order to achieve the above purpose, the present invention further provides a glue gun, including: a first handle and a second handle, where the first handle is connected to the second handle via a pivot pin shaft; one of the first handle and the second handle is arranged as a fixed handle, and the other is arranged as a movable handle rotating relative to the fixed handle; the movable handle is provided with an actuating member, and the actuating member is configured to push a push member of the glue gun; and at least one of the actuating member and the pivot pin shaft is arranged to be adjustable in position so as to switch between at least two gears, so that the ratio of a spacing between the pivot pin shaft and a force application point on the movable handle to a spacing between the pivot pin shaft and a force bearing point on the actuating member is changed, thus enabling a change in the force applied on the push member as well as a change in the rate of movement of a push rod.
(16) Further, a spacing between the axis of the actuating member and the axis of the pivot pin shaft is sized to be adjustable, such that the pivot pin shaft or the actuating member can switch between the at least two gears.
(17) Further, one of the first handle and the second handle is provided with a sliding groove, which is provided with the at least two gears; and the pivot pin shaft is arranged to be slidable in the sliding groove so as to enable switching between the gears.
(18) Further, an elastic component is further provided between the first handle and the second handle; and the elastic component is arranged to produce a pre-tightening force on one of the first handle and the second handle which is arranged as the movable handle so as to keep the pivot pin shaft in one of the gears.
(19) Further, the pivot pin shaft is a stepped pin shaft, the stepped pin shaft has a first shaft diameter portion and a second shaft diameter portion, the widths of the sliding groove at the gears are greater than the width of a communication portion of the sliding groove which communicates the gears; the shaft diameter of the first shaft diameter portion is greater than the width of the

communication portion, and the shaft diameter of the second shaft diameter portion is smaller than the width of the communication portion; and the stepped pin shaft is configured such that: the stepped pin shaft moves axially under the action of an external force, and when the first shaft diameter portion is in any of the gears, the stepped pin shaft is kept in the gear, and when the second shaft diameter portion is in the gear, the stepped pin shaft can slide along the sliding groove.
(20) Further, the stepped pin shaft has a pressing end, the pressing end is sheathed with an elastic component, and the stepped pin shaft is configured such that: under the biasing action of the elastic component, the first shaft diameter portion is in the gear; when the pressing end is pressed, the stepped pin shaft moves axially such that the second shaft diameter portion goes into the gear; and when the pressing end is released, the stepped pin shaft moves axially in the opposite direction under the action of a restoring force of the elastic component such that the first shaft diameter portion goes into the gear again.
(21) Further, one end of the stepped pin shaft is connected to a pressing portion, the pressing portion is provided with at least one positioning pin, and the axial direction of the positioning pin is parallel to the axial direction of the stepped pin shaft; the sliding groove is provided on the first handle, the second handle is provided with at least one positioning groove, and the positioning pin passes through the positioning groove and is configured to slide in the positioning groove; the positioning pin is sheathed with an elastic component, where the stepped pin shaft and the positioning pin are configured to move along with the pressing portion, and under the biasing action of the elastic component, the first shaft diameter portion is in the gear; when the pressing portion is pressed, the stepped pin shaft and the positioning pin move axially such that the second shaft diameter portion goes into the gear; and when the pressing portion is released, the positioning pin and the stepped pin shaft move axially in the opposite direction under the action of a restoring force of the elastic component such that the first shaft diameter portion goes into the gear again.
(22) Further, the glue gun further includes a blocking piece arranged opposite the pressing portion, and the other ends of the stepped pin shaft and the positioning pin are connected to the blocking piece by fasteners.
(23) Further, the pressing portion is provided with a first positioning pin and a second positioning pin, and the second handle is provided with a first positioning groove corresponding to the first positioning pin and a second positioning groove corresponding to the second positioning pin.
(24) Further, the first positioning pin, the second positioning pin and the stepped pin shaft are distributed in a triangle shape.
(25) Further, the first positioning pin and the second positioning pin are on a same straight line.
(26) Further, a boss is provided on an outer surface of the pressing portion.
(27) Further, the sliding groove is an arc-shaped groove or a linear groove.
(28) Further, the first handle is provided with a sliding groove, which is provided with the at least two gears; and the actuating part is arranged to be slidable in the sliding groove so as to enable switching between the gears.
(29) Further, the actuating member further includes a pivot, and the actuating part on the actuating member is arranged to be slidable in the sliding groove around the pivot.
(30) Further, the glue gun further includes: a brake sheathed on a push rod of the glue gun, where a compression spring is provided between the brake and a main body of the glue gun, and the brake is configured to retain the push rod under the push of the compression spring, so that the push rod can only move in the direction of glue flowing out.
(31) Further, the main body is provided with a limiting groove, and one end of the brake is positioned in the limiting groove; and the limiting groove has a first limiting end and a second limiting end, and the brake is configured to move between the first limiting end and the second limiting end, so that the push rod has an idle stroke, with a distance being that from the first limiting end to the second limiting end, in a pushing process.
(32) Further, the idle stroke is 3 to 5 mm.

(33) In order to achieve the above purpose, the present application further provides a glue gun, including: a trigger device including a movable handle and a fixed handle which are connected via a pivot pin shaft; a main body with one end formed as an accommodation part for accommodating glue and the other end connected to the fixed handle; a push rod with one end provided with a push body positioned in the accommodation part and the other end sheathed with a push member, where the push body is arranged to reciprocate along with the push rod; an actuating member provided on the movable handle and configured to push the push member; and a brake sheathed on the push rod, where a compression spring is provided between the brake and the main body, and the brake is configured to retain the push rod under the push of the compression spring, so that the push rod can only move in the direction of the glue flowing out, where a spacing between the actuating member and the pivot pin shaft is sized to be adjustable to make the glue gun switch between at least two gears, so that the ratio of a spacing between the pivot pin shaft and a force application point on the movable handle to a spacing between the pivot pin shaft and a force bearing point on the actuating member is changed, thus enabling a change in the force applied on the push member as well as a change in the rate of movement of the push rod.
(34) Further, the movable handle has a sliding groove, and the sliding groove has a first gear and a second gear; the actuating member is configured to slide in the sliding groove so as to switch between the first gear and the second gear; or the pivot pin shaft is configured to slide in the sliding groove so as to switch between the first gear and the second gear.
(35) Compared with the prior art, the beneficial effects of the present invention are as follows: 1) when the pivot pin shaft is close to the actuating part, the force of the actuating part acting on the push member is increased, which is suitable for glue with a poor fluidity; and when the pivot pin shaft is far away from the actuating part, the force of the actuating part acting on the push member is reduced, which is suitable for glue with a good fluidity; 2) the arrangement of the elastic component between the movable handle and the fixed handle can prevent the pivot pin shaft from automatically jump to other gears when a gripping force is applied on the movable handle, playing the role of fixing the gear; 3) under the biasing action of the elastic component, the stepped pin shaft can be pressed to easily realize multi-gear shifting; 4) the gear can be adjusted to enable switching between a fast feed and a powerful feed, thereby enabling the user to make adjustment according to actual requirements; and 5) the idle stroke is set such that the internal stress of the glue in a glue cylinder is released, thereby preventing the glue from flowing out of a glue outlet.
(36) The present invention further provides a glue gun, which can switch between the following two states: in a first state, after a handle is released, glue in a glue cylinder is kept dripping continuously; and in a second state, after the handle is released, the glue does not drip any more.
(37) In order to achieve this purpose, the present invention provides a glue gun, including: a main body with one end formed as an accommodation part for accommodating a glue cylinder; a fixed handle connected to another end of the main body; a movable handle pivotally connected to the fixed handle; a push rod with one end located in the accommodation part; a brake provided on the push rod; and a movable member provided on the main body, where the movable member is configured to lock the brake, so as to continuously apply a force on glue in the glue cylinder to keep an internal stress of the glue, and release the brake to release the force applied on the glue in the glue cylinder, thereby releasing the internal stress of the glue.
(38) Further, the main body is provided with a limiting groove, the limiting groove has a first limiting end and a second limiting end, one end of the brake is located in the limiting groove, and the brake is configured to move between the first limiting end and the second limiting end.
(39) Further, the movable member is configured to be movable; and the movable member has a first position and a second position, where in the first position, the movable member locks the brake, so after the movable handle is released, the internal stress of the glue is still kept, so that the glue in the glue cylinder is dripping continuously; and in the second position, the movable member unlocks the brake, so after the movable handle is released, the internal stress of the glue is released,

so that the glue does not drip any more.
(40) Further, the movable member is pivotally connected to the main body, and the movable member has an end portion facing the brake; when the movable member is in the first position, the end portion locks the brake; and when the movable member is in the second position, the end portion releases the brake.
(41) Further, a top of the main body is provided with a convex piece, the limiting groove is formed in the convex piece, and the movable member is pivotally connected to the convex piece.
(42) Further, the convex piece is provided with a pin shaft, a first through hole is provided in the movable member, and the pin shaft is sheathed in the first through hole.
(43) Further, the first through hole is located in the middle of the movable member.
(44) Further, a second through hole is provided in the movable member, a positioning boss is provided on the convex piece, and when the movable member is in the first position, a part of the positioning boss is embedded into the second through hole.
(45) Further, the positioning boss is a trapezoid boss, side faces of the trapezoid boss are inclined, and a size of a top of the trapezoid boss is smaller than a size of a bottom of the trapezoid boss.
(46) Further, the movable member is provided with a corner portion, the corner portion faces the main body, and when the movable member is in the second position, the corner portion is in contact with the main body to block the movable member.
(47) Further, the movable member is provided with an arc-shaped portion, the arc-shaped portion and the corner portion are located on a same side of the movable member, and the arc-shaped portion is always in contact with the main body.
(48) Further, the movable member includes a first part and a second part that are the same and are symmetrically arranged, there is a gap between the first part and the second part, and the convex piece is inserted into the gap; and the first part and the second part are connected together by a connection portion.
(49) Further, the connection portion is provided with a proximal end and a distal end, and the movable member is configured such that when the proximal end is pressed, the movable member is rotated to the first position, and when the distal end is pressed, the movable member is rotated to the second position.
(50) Further, a rotating shaft of the movable member is provided at an end away from the brake.
(51) Further, the movable member is configured such that when an end of the movable member opposite to the rotating shaft is pressed, the movable member is moved to the first position.
(52) Further, the movable member is provided on one side of the convex piece, and a rotating shaft of the movable member is provided at an end of the movable member away from the brake.
(53) Further, the rotating shaft of the movable member includes a screw passing through the convex piece and a nut provided at one end of the screw, and the movable member is provided with a sleeve with which the nut is sheathed.
(54) Further, a blind hole is provided in the movable member, the blind hole and the sleeve are provided on a same side of the movable member, the convex piece is provided with a protrusion, and an elastic element connects the protrusion and the blind hole respectively.
(55) Further, a groove is provided in the movable member, a blocking portion is protruded on the convex piece, and the blocking portion falls into the groove when the movable member is in the second position.
(56) Further, the blocking portion is obliquely provided on the convex piece.
(57) The concept, specific structures, and technical effects of the present invention will be further illustrated below in conjunction with accompanying drawings, such that the purpose, features, and effects of the present invention can be fully understood.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram showing the structure of a glue gun according to Embodiment 1 of the present invention;

(2) FIG. **2** is a schematic exploded diagram showing a partial structure of the glue gun in FIG. **1**;

(3) FIG. **3** is a schematic diagram of a first gear in FIG. **1**;

(4) FIG. **4** is a schematic diagram of a second gear in FIG. **1**;

(5) FIG. **5** is a schematic diagram showing the structure of a glue gun according to Embodiment 2 of the present invention;

(6) FIG. **6** is a schematic exploded diagram showing a partial structure of the glue gun in FIG. **5**;

(7) FIG. **7** is an A-A section view of the glue gun in FIG. **5**;

(8) FIG. **8** is a schematic diagram showing the structure of a stepped pin shaft in FIG. **5**;

(9) FIG. **9** is a schematic diagram showing the structure of a linear sliding groove in FIG. **5**;

(10) FIG. **10** is a schematic diagram showing the structure of a glue gun according to Embodiment 3 of the present invention;

(11) FIG. **11** is a schematic diagram showing a partial structure in FIG. **10**;

(12) FIG. **12** is a section view of an actuating member in FIG. **10**;

(13) FIG. **13** is a schematic diagram showing the structure of a glue gun according to Embodiment 4 of the present invention;

(14) FIG. **14** is a rear view of the part of FIG. **13**;

(15) FIG. **15** is a schematic exploded diagram of FIG. **13**;

(16) FIG. **16** is a partially enlarged diagram of FIG. **13**;

(17) FIG. **17** is a schematic diagram from another perspective of FIG. **15**;

(18) FIG. **18** is a C-C section view of FIG. **13**;

(19) FIG. **19** is a D-D section view of FIG. **13**;

(20) FIG. **20** is a schematic diagram showing a connection between a brake and a main body in Embodiments 1 to 4;

(21) FIG. **21** is a schematic diagram showing another connection between a brake and a main body in Embodiments 1 to 4;

(22) FIG. **22** is a schematic diagram showing the structure of a glue gun in a first state according to Embodiment 5;

(23) FIG. **23** is a schematic partially-enlarged diagram of FIG. **22**;

(24) FIG. **24** is a schematic diagram showing the structure of a glue gun in a second state according to Embodiment 5;

(25) FIG. **25** is a schematic partially-enlarged diagram of FIG. **24**;

(26) FIG. **26** is a schematic partially-exploded diagram of a glue gun according to Embodiment 5;

(27) FIG. **27** is a schematic diagram showing the structure of a positioning boss of a glue gun according to Embodiment 5;

(28) FIG. **28** is a schematic partially-enlarged diagram of a glue gun in a second state according to Embodiment 5;

(29) FIG. **29** is a schematic diagram showing the structure of a glue gun according to Embodiment 5 from another perspective;

(30) FIG. **30** is a schematic diagram showing the structure of a glue gun in a first state according to Embodiment 6;

(31) FIG. **31** is a schematic diagram showing the structure of a glue gun in a second state according to Embodiment 6;

(32) FIG. **32** is a schematic partially-enlarged diagram of a glue gun in a first state according to Embodiment 7;

(33) FIG. **33** is a schematic structural diagram of a reverse side of the FIG. **32**;

(34) FIG. **34** is a schematic axonometric diagram of FIG. **32**;

(35) FIG. **35** is a schematic exploded diagram of FIG. **32**;
(36) FIG. **36** is a schematic diagram from another perspective of FIG. **35**; and
(37) FIG. **37** is a schematic diagram from another perspective of FIG. **35**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(38) A plurality of preferred embodiments of the present invention will be described hereafter with reference to the accompanying drawings of the description, so that the technical contents thereof will be more clearly and easily understood. The present invention can be embodied by many different forms of examples, and the protection scope of the present invention is not limited to the examples mentioned in the text.

(39) In the drawings, components with the same structure are denoted by the same reference numeral, and components with similar structures or functions are denoted by similar reference numerals. The size and thickness of each constituent part as shown in the drawings are arbitrarily shown, and the present invention does not limit the size and thickness of each constituent part. In order to make the illustration clearer, the thickness of parts is appropriately exaggerated somewhere in the drawings.

Embodiment 1

(40) FIGS. **1** to **4** show a preferred embodiment of the present invention, and as shown in FIGS. **1** and **2**, a glue gun of this embodiment includes a push device, a main body **3** and a trigger device. An accommodation part **1** is formed at one end of the main body **3**, and the accommodation part **1** is provided in the shape of a cylinder so as to accommodate a glue cylinder. The other end of the main body **3** and the trigger device are hinged via a fastener or are integrally formed to form a gun-shaped fixed handle **16**.

(41) The push device includes a push member **8**, a push rod **2** and a push body **10**. A first end of the push rod **2** is placed in the accommodation part **1**. One end of the accommodation part **1** is connected to the main body **3**, and the other end is provided with an outlet for a nozzle of the glue cylinder to pass through. The push body **10** is fixed to an end portion of the first end of the push rod **2** and can reciprocate along with the push rod **2**. A second end of the push rod **2** is sheathed with the push member **8**, and a restoring spring **7** is also provided between the push member **8** and the main body **3**. The trigger device pushes the push member **8**, so that the push rod **2** and the push body **10** moves together towards the outlet of the accommodation part **1**, and the restoring spring **7** restores the push member **8** in position. The push rod **2** is also provided with a brake **4**, and a compression spring **9** is provided between the brake **4** and the main body **3**. The brake **4** retains the push rod **2** under the push of the compression spring **9**, so that the push rod **2** can only move towards the accommodation part **1**. When the glue cylinder needs to be installed, the brake **4** is pressed to release the push rod **2**, so as to adjust the position of the push rod **2**. The brake **4** cooperates with a limiting groove **31** of the main body **3**, and one end of the brake **4** moves between a first limiting end **32** and a second limiting end **33** of the limiting groove **31**. When the brake **4** is located at the second limiting end **33**, the brake **4** retains the push rod **2** under an elastic force of the compression spring **9**. In glue feeding, the push member **8** is pushed to move toward the outlet of the accommodation part **1**, the push rod **2** and the brake **4** will accordingly move together, and the brake **4** moves from the second limiting end **33** to the first limiting end **32**. There is no relative displacement between the push rod **2** and the brake **4**, that is, the brake **4** has an idle stroke. When the brake **4** moves to the first limiting end **32**, the brake **4** is blocked by the first limiting end **32** and does not move any more. At this time, the brake **4** will not retain the push rod **2**, so that the push rod **2** can continue to move towards the accommodation part **1**, and in this case, a relative displacement occurs between the push rod **2** and the brake **4**. The idle stroke is the distance between the first limiting end **32** and the second limiting end **33**, and preferably, the length of the idle stroke is 3 to 5 mm. When the glue feeding is finished, a movable handle **5** is released. Under the action of the restoring spring **7**, the push member **8** will move away from the accommodation part **1**, driving the push rod **2** to move together. When the push rod **2** moves to a

position where it is retained by the brake **4**, the brake **4** will also move together with the push rod **2**. At this time, the brake **4** will move from the first limiting end **32** to the second limiting end **33** so as to release the force acting on the glue inside the glue cylinder, so that the internal stress of the glue inside the glue cylinder is released, thereby preventing the glue from flowing out of a glue outlet. When the brake **4** moves to the second limiting end **33**, the brake **4** locks the push rod **2** under the action of the compression spring **9**. At this time, the push member **8** will continue to move away from the push rod **2** until the movable handle **5** is restored in position. Repeat the above actions to continue the glue feeding.

(42) The trigger device includes the movable handle **5** and the fixed handle **16**. The movable handle **5** is connected to the fixed handle **16** via a pivot pin shaft **12**. The movable handle **5** is further provided with an actuating member **11**, and the actuating member **11** passes through a hole **14** in the movable handle **5** and makes contact with the push member **8**. The movable handle **5** is provided with a sliding groove **13**, and the sliding groove **13** is of an arc-shaped structure and has two gears, that is, a first gear **131** and a second gear **132**. The first gear **131** and the second gear **132** may be respectively located at two end portions of the sliding groove **13**, and the widths of the sliding groove **13** at the first gear **131** and the second gear **132** are greater than the width of other part of the sliding groove **13**. The pivot pin shaft **12** can slide in the sliding groove **13**. When the pivot pin shaft **12** slides to a position of the first gear **131** of the sliding groove **13**, a tension spring **6** is provided between the movable handle **5** and the fixed handle **16**. One end of the tension spring **6** is connected to a tension spring groove **15** of the movable handle **5** and the other end is connected to the fixed handle **16**. The tension spring **6** produces a pre-tightening force on the movable handle **5** so as to keep the pivot pin shaft **12** in one of the gears. When a gripping force is applied to the movable handle **5**, the pivot pin shaft **11** pushes a pushing surface **17** of the push member **8** to move the push rod **2**. When the applied force is removed, the brake **4** locks the push rod **2**, and under the action of the restoring spring **7**, the push member **8** slides and restores to an initial position relative to the main body **3**, so that the next cycle can be carried out.

(43) FIGS. **3** and **4** show the variation of the spacing between the pivot pin shaft **12** and the actuating member **11** in different gears. When the pivot pin shaft **12** is in the first gear **131**, the spacing between the pivot pin shaft **12** and the actuating member **11** is set as L3. Here, the spacing refers to a distance between the axis of the pivot pin shaft **12** and the axis of the actuating member **11** (In FIGS. **3** and **4**, the axis of the pivot pin shaft **12** and the axis of the actuating member **11** are both perpendicular to the plane of the paper and facing outward, that is, on a side face **51** of the movable handle **5**, indicating a distance between the center of the circle of the pivot pin shaft **12** and the center of the circle of the actuating member **11**). The vertical spacing between the pivot pin shaft **12** and the force applied on the movable handle **5** is set as L4. This vertical spacing refers to a distance between the axis of the pivot pin shaft **12** and a force application point **52** on the movable handle **5** in a vertical direction Y. When the pivot pin shaft **12** is in the second gear **132**, the spacing between the pivot pin shaft **12** and the actuating member **11** is set as L1, and the vertical spacing between the pivot pin shaft **12** and the force applied on the movable handle **5** is set as L2. The position of the pivot pin shaft **12** relative to the actuating member **11** changes when it is in different gears, so $L1>L3$. However, no matter which gear the pivot pin shaft **12** is in, its position is unchanged, so $L2=L4$. When the gear of the pivot pin shaft **12** is adjusted to the second gear **132**, a force F is applied on the movable handle **5**, and a force applied by the actuating member **11** on the push member **8** is F1, then $F*L2=F1*L1$; and when the gear of the pivot pin shaft **12** is adjusted to the first gear **131**, a same force F is applied on the movable handle **5**, and a force applied by the actuating member **11** on the push member **8** is F2, then $F*L4=F2*L3$. It can be seen that when the same force F is applied on the movable handle, since $L1>L3$, there is $F1<F2$, that is, when in the first gear **131**, the pushing force generated by applying the same force on the movable handle **5** is greater than that when in the second gear **132**. In the second gear **132**, when the movable handle **5** moves, it moves with the pivot pin shaft **12** as a circular point, and the actuating member **11** moves

in an arc with the length of L1 as the radius to push the push member **8** to move forward. In the first gear **131**, when the movable handle **5** moves, it moves with the pivot pin shaft **12** as a circular point, and the actuating member moves in an arc with the length of L3 as the radius to push the push member **8** to move forward. Since L1>L3, when the same force F is applied on the movable handle, the push member is pushed forward a further distance in the gear **132**. Therefore, in the gear **132**, the glue gun is suitable for glue with lower viscosity, and a longer pushing course can be achieved with a smaller pushing force. By changing the gear of the pivot pin shaft **12**, the spacing between the pivot pin shaft **12** and the actuating member **11** is changed, such that the pushing force acting on the push member **8** by the actuating member **11** is different, so as to change the pushing course of the push rod **2**, so it can adapt to fluid with a different fluidity. In other words, in this embodiment, by adjusting the ratio of the spacing between the pivot pin shaft **12** and the force application point **52** to the spacing between the pivot pin shaft **12** and a force bearing point (i.e., a point where the actuating member **11** contacts the push member), the adjustment of the force applied on the push member and the switching between transmission rates (i.e., speeds at which the push rod is pushed) are realized.

(44) In other embodiments, the sliding groove **13** may be provided on the fixed handle **5**, and the sliding groove **13** is provided with two or more gears. The pivot pin shaft **12** is arranged to be slidable in the sliding groove **13** so as to enable switching between the gears **131** and **132**.

Embodiment 2

(45) In other embodiments, as shown in FIGS. **5** to **9**, the tension spring **6** between the movable handle **5** and the fixed handle **16** is removed. The pivot pin shaft **12** is provided as a stepped pin shaft **102**, and the stepped pin shaft **102** has a pressing end and an end fixed to a screw **101**. The pressing end is sheathed with a spring **104** and a spacer **105**. The stepped pin shaft **102** has a first shaft diameter portion **107** and a second shaft diameter portion **108** (see FIG. **8**). The sliding groove **103** has a first gear **1031** and a second gear **1032**. The first gear **1031** and the second gear **1032** may be respectively located at two end portions of the sliding groove **103**. The widths of the sliding groove **103** at the first gear **1031** and the second gear **1032** are greater than the width of other part (i.e., a middle portion connecting the first gear **1031** and the second gear **1032**) of the sliding groove **103**. The stepped pin shaft **102** may slide in the sliding groove **103**, so as to move to the first gear **1031** or the second gear **1032**. The shaft diameter of the first shaft diameter portion **107** is greater than the width of the middle portion of the sliding groove **103**, and the shaft diameter of the second shaft diameter portion **108** is smaller than the width of the sliding groove **103**. Under the biasing action of the spring **104**, the first shaft diameter portion **107** is in the gear **1031** or **1032**, and when the pressing end is pressed, the stepped pin shaft **102** moves axially such that the second shaft diameter portion **1032** goes into the gear **1031** or **1032**, and at the same time, the shaft diameter of the second shaft diameter portion **1032** is smaller than the width of the sliding groove **103**, such that the stepped pin shaft **102** may be moved to slide along the sliding groove **103**, thus making the stepped pin shaft **102** move from one gear to another for gear switching. When the pressing end is released, the stepped pin shaft **102** moves axially in the opposite direction under the action of a restoring force of a spring **104**, such that the first shaft diameter portion **107** goes into the current gear again. Since the shaft diameter of the first shaft diameter portion **107** is greater than the width of the sliding groove **103**, the stepped pin shaft **102** is fixed at the current position of the sliding groove **103**, thereby fixing the stepped pin shaft **102** in the current gear. Preferably, as shown in FIG. **9**, the sliding groove **301** is linear and has multiple gears **3011**, **3012** and **3013**. The widths of the sliding groove **301** at the gears are greater than the widths of other parts (i.e. parts between adjacent gears). The stepped pin shaft **302** slides in the sliding groove **301** to realize the switching between the gears. In this embodiment, by changing the position of the stepped pin shaft in the sliding groove, the stepped pin shaft can be in different gears. In this case, the spacing between the stepped pin shaft and the actuating member **11** is changed, so the vertical spacing between the stepped pin shaft and the force application point on the movable handle **5** is changed,

such that the pushing force acting on the push member **8** by the actuating member **11** is different, so as to change the pushing course of the push rod **2**, so it can adapt to fluid with a different fluidity. In other words, in this embodiment, by adjusting the ratio of the spacing between the stepped pin shaft and a force application point to the spacing between the stepped pin shaft and a force bearing point (i.e., a point where the actuating member **11** contacts the push member), the adjustment of the force applied on the push member and the switching between transmission rates (i.e., speeds at which the push rod is pushed) are realized.

Embodiment 3

(46) FIGS. **10** to **12** show another preferred embodiment of the present invention. The movable handle **5** is hinged to the fixed handle **16** by means of a pivot pin shaft **201**. The movable handle **5** is further provided with a sliding groove **202**. The actuating member further includes a pivot **204** and an actuating part **206** in contact with the push member **8**. The actuating part **206** is arranged to be slidable in the sliding groove **202** around the pivot **204**. The sliding groove **202** has a first gear **2021** and a second gear **2022**. The first gear **2021** and the second gear **2022** may be respectively located at two ends of the sliding groove **202**. A shift member **203** is connected to the pivot **204** and the actuating part **206**, and the actuating part **206** slides between two ends of the sliding groove **202** by means of the shift member **203**, i.e. realizing the changes in the position where the actuating part **206** makes contact with the push member **8** to achieve switching between the gears **2021** and **2022**. When the actuating part **206** is in different gears, the spacing between the actuating part **206** and the pivot pin shaft **201** is different, and the vertical spacing between the pivot pin shaft **201** and a force application point on the movable handle **5** is different, such that the pushing force acting on the push member **8** by the actuating part **206** is different, so as to change the pushing course of the push rod **2**, so it can adapt to fluid with a different fluidity. In Embodiment 1 and Embodiment 2, by changing the gear of the pivot pin shaft, the purpose of changing the spacing between the pivot pin shaft and the actuating member is achieved; while in this embodiment, the change in the spacing between the actuating part and the pivot pin shaft is achieved by changing the gear of the actuating part **206**. In other words, in this embodiment, by adjusting the ratio of the spacing between the pivot pin shaft **201** and a force application point to the spacing between the pivot pin shaft **201** and a force bearing point (i.e., a point where the actuating part **206** contacts the push member **8**), the adjustment of the force applied on the push member and the switching between transmission rates (i.e., speeds at which the push rod is pushed) are realized.

Embodiment 4

(47) FIGS. **13** to **19** show another preferred embodiment of the present invention. Referring to FIG. **13**, most features of this embodiment are the same as those of Embodiment 1. For example, components such as the push member **8**, the push rod **2**, the push body **10**, the main body **3**, the accommodation part **1**, the brake **4**, the movable handle **5**, the restoring spring **7**, the compression spring **9**, the actuating member **11** and the fixed handle **16**, and their connection modes are the same as those in Embodiment 1, which will not be repeated here. This embodiment is different from Embodiment 1 in that the way of shifting gears of the pivot pin shaft is different.

(48) Referring to FIG. **15**, in this embodiment, the actuating member **11** passes through a hole **14** in the movable handle **5** and makes contact with the push member **8** to drive the push member **8**, so as to move the push rod **2**. The movable handle **5** is connected to the fixed handle **16** via a pivot pin shaft **401**. When a force is applied to the movable handle **5**, the movable handle **5** can rotate around the pivot pin shaft **401** relative to the fixed handle **16**. The movable handle **5** is provided with a sliding groove **402**. Referring to FIG. **16**, the sliding groove **402** has a first gear **4021** and a second gear **4022**. The first gear **4021** and the second gear **4022** can be respectively arranged at both ends of the sliding groove **402**, that is, holes are formed at both ends of the sliding groove **402** as the first gear **4021** and the second gear **4022**, and the two holes communicate with each other to form a communication portion **4023**, thus forming the sliding groove **402**, where the diameter of the holes is greater than the width of the communication portion **4023**. The pivot pin shaft **401** can slide in

the sliding groove **402** to move to the first gear **4021** or the second gear **4022**. Referring to FIGS. **15** and **19**, the pivot pin shaft **401** is of a stepped pin shaft structure, including a first shaft diameter portion **4011** and a second shaft diameter portion **4012**. The shaft diameter of the first shaft diameter portion **4011** is greater than the width of the communication portion **4023** of the sliding groove **402**, and the shaft diameter of the second shaft diameter portion **4012** is smaller than the width of the communication portion **4023** of the sliding groove **402**. When the first shaft diameter portion **4011** is in the first gear **4021** or the second gear **4022** of the sliding groove **402**, the pivot pin shaft **401** cannot slide along the sliding groove **402**, thus keeping the pivot pin shaft **401** in the current gear. When the second shaft diameter portion **4012** is located in the sliding groove **402**, since the shaft diameter of the second shaft diameter portion **4012** is smaller than the width of the sliding groove **402**, the fixing of the pivot pin shaft **401** can be released, such that the pivot pin shaft **401** can slide along the sliding groove **402** under the action of an external force to switch gears.

(49) Referring to FIGS. **15** and FIGS. **17** to **19**, one end of the pivot pin shaft **401** is connected to a pressing portion **403**, the pressing portion **403** is further provided with at least one positioning pin **404**, the axial direction of the positioning pin **404** is parallel to the axial direction of the pivot pin shaft **401**, the fixed handle **16** is provided with at least one positioning groove **405**, one positioning pin **404** passes through one positioning groove **405**, and the positioning pin **404** can slide along the positioning groove **405** in a sliding direction substantially same as that of the pivot pin shaft **401** in the sliding groove **402**. The positioning pin **404** is sheathed with a spring **406**. When the pressing portion **403** is pressed, the pivot pin shaft **401** and the positioning pin **404** both move along their respective axial directions, the second shaft diameter portion **4012** of the pivot pin shaft **401** moves into the sliding groove **402**, and the spring **406** over the positioning pin **404** is compressed. Then the pressing portion **403** is driven to move along the length direction of the sliding groove **402**, the pivot pin shaft **401** slides in the sliding groove **402**, and the positioning pin **404** slides in the positioning groove **405**. When the pivot pin shaft **401** is switched to a preset gear, the pressing portion **403** is released, and under the biasing action of the spring **406**, the pressing portion **403** is restored in position, and both the pivot pin shaft **401** and the positioning pin **404** move along with the pressing portion **403**. At this time, the first shaft diameter portion **4011** of the pivot pin shaft **401** moves to the gear **4021** or **4022** of the sliding groove **402**, so that the pivot pin shaft **401** is kept in the current gear.

(50) Preferably, the number of the positioning pins **404** are two. Accordingly, the fixed handle **16** is provided with two positioning grooves **405** corresponding to the positioning pins **404**, each positioning pin **404** being sheathed with a spring **406**. The two positioning pins **404** and the pivot pin shaft **401** form a triangle, where the two positioning pins **404** may be on a same straight line, the direction of which may be the same as the length directions of the positioning grooves **405**. The shape of the pressing portion **403** may be substantially triangular.

(51) A blocking piece **409** is provided on a side opposite the pressing portion **403**, and the other ends of the pivot pin shaft **401** and the positioning pins **404** are all connected to the blocking piece **409** by fasteners **407** such as screws.

(52) Marks **408** are provided on the movable handle **5** to indicate that the pivot pin shaft **401** is in different gears. A boss **4031** is provided on an outer surface of the pressing portion **403**, which is convenient for a user to operate the pressing portion **403** to press the pressing portion **403** and push the pressing portion **403**.

(53) Similar to Embodiments 1 to 3, in this embodiment, when the pivot pin shaft **401** is in different gears in the sliding groove **402**, the spacing between the pivot pin shaft **401** and the actuating member **11** (i.e., the distance between the axis of the pivot pin shaft **401** and the axis of the actuating member **11** on a side face **51** of the movable handle **5**) is different, and the vertical spacing between the pivot pin shaft **401** and a force application point **52** on the movable handle **5** is different, such that the contact position of the actuating member **11** and the push member **8** is

different and the pushing force acting on the push member **8** is different, so as to change the pushing course of the push rod **2**, so it can adapt to fluid with a different fluidity. In other words, in this embodiment, by adjusting the ratio of the spacing between the pivot pin shaft **401** and a force application point to the spacing between the pivot pin shaft **401** and a force bearing point (i.e., a point where the actuating member **11** contacts the push member), the adjustment of the force applied on the push member and the switching between transmission rates (i.e., speeds at which the push rod is pushed) are realized.

Embodiment 5

(54) In Embodiments 1 to 4, referring to FIG. **20**, an upper portion of the main body **3** close to the brake **4** is provided with the limiting groove **31**, and one end (a top end **41**) of the brake **4** moves in the limiting groove **31**. Under the constraints of the first limiting end **32** and the second limiting end **33** of the limiting groove **31**, in the glue feeding, the brake **4** moves from the second limiting end **33** to the first limiting end **32**; and after the glue feeding is finished, the handle is released, and the brake **4** will move from the first limiting end **32** to the second limiting end **33**. Of course, the first limiting end **32** may also be removed, and an end portion **34** of the main body **3** facing the brake **4** is taken as the limiting end (see FIG. **21**). In the above two ways, no matter which way it is, the brake **4** has an idle stroke during the movement process, that is, during the glue feeding process, the actuating member **4** will move together with the push rod **2**, and there is no relative displacement between them. When the glue gun is not working, after the handle is released, the brake **4** will move backwards (in an X direction) together with the push rod **2** (see the direction X in FIG. **20**), thus releasing the force acting on the glue to release the internal stress of the glue. At this time, the glue in the glue cylinder can be prevented from dripping any more.

(55) However, in some application scenarios, it is desirable that after the handle is released, a continuous force is still applied on the glue in the glue cylinder, so that the glue can be dripping continuously. At this time, it is needed that after the handle is released, the brake **4** can be locked, so that the brake **4** will not move with the release of the handle, so as to keep it in a force-applying state and make the glue dripping continuously.

(56) Meanwhile, whether to lock the brake **4** can be selected according to user requirements, that is, the brake **4** is set to switch between a locked state and a movable state. In order to solve this problem, this embodiment is improved on the basis of Embodiments 1 to 4, and the detailed description is as follows.

(57) Referring to FIGS. **22** to **25**, the arrangement of the brake **4** is the same as that of Embodiments 1 to 4, that is, the brake **4** is arranged on the push rod **2**, and the compression spring **9** is arranged between the brake **4** and the main body **3**, the top end of the brake **4** cooperates with the limiting groove **31** of the main body **3**, and the top end of the brake **4** can move back and forth in the limiting groove **31** (the forward direction is the Y direction, and the backward direction is the X direction). Referring to FIGS. **22** and **23**, a movable member **500** is provided at the top of the main body **3**, and an end portion **501** of the movable member **500** facing the brake **4** may be moved under the action of an external force. When the end portion **501** moves to a position (a first position) where it is in contact with the top end **41** of the brake **4** (or there is a very small gap between the end portion **501** and the top end **41**), the end portion **501** locks the brake **4** together with the second limiting end **33**, such that the brake **4** cannot move in the limiting groove **31**, and the brake **4** will retain the push rod **2** and prevent the push rod **2** from moving. At this time, the glue gun is in a first state, that is, after the handle is released, the restoring force of the restoring spring **7** is not enough to overcome the force of the brake **4** on the push rod **2**, such that the brake **4** remains in a state of retaining the push rod **2**. The push rod **2** does not move, a force is still applied to the glue, and the internal stress of the glue is not released, so that the glue is still in a state of continuous dripping.

(58) As shown in FIGS. **24** and **25**, a user applies an external force on the movable member **500** to move the end portion **501** of the movable member **500** to a position (a second position) away from

the brake **4**. At this time, the end portion **501** of the movable member **500** is no longer in contact with the top end of the brake **4**, and in the entire movement stroke of the brake **4**, the end portion **501** of the movable member **500** will not be in contact with the brake **4**, that is to say, the movable member **500** will not interfere with the movement of the brake **4** in the limiting groove **31**. The top end **41** of the brake **4** can move back and forth in the limiting groove **31**, that is, the top end **41** of the brake **4** can move between the end portion **34** of the main body **3** and the second limiting end **33**. When the brake **4** is located at the second limiting end **33**, the brake **4** retains the push rod **2**. In the glue feeding, the push member **8** is pushed to move toward the accommodation part **1**, the push rod **2** and the brake **4** will accordingly move together, and the brake **4** moves from the second limiting end **33** to the end portion **34**. There is no relative displacement between the push rod **2** and the brake **4**, that is, the brake **4** has an idle stroke. When the brake **4** moves to the end portion **34**, the brake **4** is blocked by the end portion **34** and does not move any more. At this time, the brake **4** will not retain the push rod **2**, so that the push rod **2** can continue to move towards the accommodation part **1**, and in this case, a relative displacement occurs between the push rod **2** and the brake **4**. When the movable member **500** is in the second position, the glue gun is in a second state, that is, under the action of the restoring spring **7**, the push member **8** will move away from the accommodation part, driving the push rod **2** to move together. When the push rod **2** moves to a position where it is retained by the brake **4**, the brake **4** will also move together with the push rod **2**. At this time, the brake **4** will move from the end portion **34** to the second limiting end **33** so as to release the force acting on the glue inside the glue cylinder, so that the internal stress of the glue inside the glue cylinder is released, thereby preventing the glue from flowing out of a glue outlet, that is, the glue does not drip any more.

(59) The movable member **500** may adopt any suitable structure, as long as it can make the end portion of the movable member move to the first position (the position where the end portion is in contact with the top end **41** of the brake **4** and locks the brake **4**) and move to the second position (the position where the end portion is out of contact with the top end **41** of the brake **4** and does not interfere with the movement of the brake **4**), and any movable member meeting this requirement can be applied in this embodiment.

(60) As shown in FIGS. **22** to **29**, this embodiment provides a preferred implementation, as follows: Referring to FIG. **26**, a convex piece **35** is provided at the top of the main body **3**. The convex piece **35** extends upward and obliquely rearward from the top of the main body **3**. An end portion of the convex piece **35** away from the main body **3** is formed as a second limiting end **33**. The second limiting end **33** and the end portion **34** of the main body **3** facing the brake **4** form the limiting groove **31** in an enclosing manner. The top end **41** of the brake **4** can move back and forth in the limiting groove **31**. The movable member **500** is pivotally connected to the convex piece **35**. Specifically, the convex piece **35** is provided with a pin shaft **351**. A first through hole **502** is provided in the movable member **500**. The pin shaft **351** is sheathed in the first through hole **502**, so that the movable member **500** can rotate around the pin shaft **351**. When the movable member **500** is rotated to the first position, the end portion **501** of the movable member **500** is in contact with the top end **41** of the brake **4** (or there is a very small gap between the top end **41** of the brake **4** and the end portion **501**), thereby locking the brake **4**. When the movable member **500** is rotated to the second position, the end portion **501** of the movable member is away from the brake **4**, so that the brake **4** can move in the limiting groove **31**, and its movement stroke is restricted by the end portion **34** of the main body **3** and the second limiting end **33**.

(61) Preferably, the first through hole **502** of the movable member **500** is located in the middle of the movable member **500**.

(62) An end face **503** of the movable member **500** facing the main body **3** may be matched with the main body **3** in shape. In this way, when the movable member **500** moves to the first position, the end face **503** of the movable member **500** is in contact with the main body **3**, and because of their matching shapes, they are in closer contact (see FIG. **23**). With this arrangement, the main body **3**

can better restrict the movement of the movable member **500**, so that the movable member **500** can move to the first position precisely. For example, the end portion **34** of the main body **3** close to the brake **4** is provided as an inclined face (the inclined face and the second limiting end **33** together form the limiting groove **31**), and the end face **503** of the movable member **500** is also provided as an inclined face matching with that inclined face.

(63) In order to better position the movable member **500** in the first position, the movable member **500** is further provided with a second through hole **504**, and correspondingly, a positioning boss **352** is provided on the convex piece **35**. When the movable member **500** moves to the first position, the second through hole **504** faces the positioning boss **352**, so that a part of the positioning boss **352** may be embedded into the second through hole **504** and the movable member **500** can be accurately positioned in the first position, and at the same time, the movable member **500** can also be kept in the first position. Referring to FIG. **27**, the positioning boss **352** is formed by protruding outward from a side face of the convex piece **35**, and the size of its bottom is larger than that of its top, such that side faces **3521** of the positioning boss **352** are inclined, which makes the matching between the positioning boss **352** and the second through hole **504** smoother. The shape of the positioning boss **352** on a cross-section parallel to the side face of the convex piece **35** where it is located may be provided to be substantially trapezoid, that is, a top face **3522** of the positioning boss **352** is substantially trapezoid, and its bottom is also substantially trapezoid. The positioning boss **352** is a trapezoid boss. The cross-sectional shape of the corresponding second through hole **504** is matched with the positioning boss **352**.

(64) Referring to FIGS. **25** and **26**, in order to better position the movable member **500** in the second position, a blocking portion **36** is provided on the main body **3**, the blocking portion **36** is located at an end of the convex piece **35** away from the brake **4**, and a side face of the blocking portion **36** facing the convex piece **35** is provided as an inclined face. When the movable member **500** is rotated to the second position, an end portion **505** of the movable member **500** away from the brake **4** is in contact with the inclined face of the blocking portion **36**, thereby accurately positioning the movable member **500** in the second position. In another implementation, a corner portion **506** may be provided on the movable member **500**. When the movable member **500** moves to the second position, the corner portion **506** is in contact with the top of the main body **3** to form a block, thereby accurately positioning the movable member **500** in the second position. Preferably, the end face of the movable member **500** facing the main body **3** is provided with an arc-shaped portion **507**. The arc-shaped portion **507** is located below the first through hole **502**. The corner portion **506** is provided on a side of the arc-shaped portion **507** away from the brake **4**, and the shape of a side close to the brake **4** matches the shape of the main body **3**. The arc-shaped portion **507** is always in contact with the top of the main body **3**, which is equivalent to that the arc-shaped portion **507** forms a fulcrum. In another implementation, referring to FIG. **28**, a top **353** of the convex piece **35** is provided as an inclined face, so that when the movable member **500** is in the first position, there is a gap **354** between the end portion **505** of the movable member **500** away from the brake **4** and the top of the convex piece **35**, and when the movable member **500** moves to the second position, the end portion **505** of the movable member **500** away from the brake **4** is in contact with the top of the convex piece **35** to form a block, thereby accurately positioning the movable member **500** in the second position.

(65) The movable member **500** may be provided only on one side of the convex piece **35**, or may be provided on both opposite sides of the convex piece **35**. As shown in FIG. **29**, the convex piece **35** is in a shape of a sheet. The movable member **500** includes a first part **510** and a second part **520** that are symmetrical. The shape of the first part **510** and the shape of the second part **520** are exactly the same. That is, the first part **510** is provided with the through hole **502**, the second through hole **504**, the corner portion **506**, the arc-shaped portion **507**, etc., and the second part **520** has the same configuration. The first part **510** is located on one side of the convex piece **35**, and the second part **520** is located on the other side of the convex piece **35**. A gap **511** is formed between

the first part **510** and the second part **520** into which the convex piece **35** may be inserted. The first part **510** and the second part **520** are both provided with the first through holes **504**, and correspondingly, the pin shafts **351** are provided on both sides of the convex piece **35**. The two pin shafts **351** are respectively located in the first through hole **504** of the first part **510** and the first through hole **504** of the second part **520**. The first part **510** and the second part **520** are connected together by a connection portion **530**, so that the movable member **500** is integrated. In this way, by pressing one end (a proximal end **532**) of the connection portion **530** facing the brake **4**, the movable member **500** is rotated to the first position; and by pressing one end (a distal end **531**) of the connection portion **530** away from the brake **4**, the movable member **500** is rotated to the second position.

Embodiment 6

(66) As shown in FIGS. **30** and **31**, most of the structures in this embodiment are the same as those in Embodiment 5, and the only difference is that, in this embodiment, a rotating shaft **601** (i.e., a pin shaft provided on the convex piece) of a movable member **600** is provided at an end of the movable member away from the brake **4**. In this embodiment, when operating the movable member, the movable member **600** can be moved to a first position only by pressing an end opposite to the rotating shaft **601**, and the movable member **600** can be moved to a second position by reverse operation.

Embodiment 7

(67) As shown in FIGS. **32** to **37**, the difference between this embodiment and Embodiment 5 is that the structure of a movable member **700** is different.

(68) Referring to FIG. **32**, the movable member **700** is provided on one side of the convex piece **35**, and an end of the movable member **700** away from the brake **4** is provided with a rotating shaft. By operating an end portion **701** of the movable member **700** close to the brake **4**, the movable member **700** may be rotated. When the movable member **700** is in a first position, the end portion **701** is in contact with the brake **4**, thereby locking the brake **4**. When the movable member **700** is in a second position, the end portion **701** is away from the brake **4** and no longer blocks the movement of the brake **4**.

(69) Referring to FIGS. **34** to **37**, the rotating shaft includes a screw **702**, and a nut **703** provided at an end of the screw **702**. The screw **702** passes through the convex piece **35**, and the screw **702** is sheathed with the nut **703**. A sleeve **704** is provided on a side of the movable member **700** facing the convex piece **35**, and the nut **703** is sheathed with the sleeve **704**. When the movable member **700** is rotated, the sleeve **704** rotates around the screw **702** together with the nut **703**. Through the cooperation of the screw **702** and the nut **703**, the movable member **700** may be kept in a current position.

(70) A blind hole **706** is provided in the side of the movable member **700** facing the convex piece **35**. Correspondingly, a protrusion **355** is provided on the convex piece **35**. The protrusion **355** is sheathed with a spring **705**, and the other end of the spring **705** is inserted into the blind hole **706**, so that the spring **705** applies an elastic force on the movable member **700**. By pressing the movable member **700** towards the convex piece **35**, the movable member **700** can be rotated against the elastic force of the spring **705**. After releasing, under the action of the elastic force, the movable member **700** moves away from the convex piece **35**, so that the movable member **700** can be locked in the current position.

(71) A groove **707** is provided in an upper portion of the movable member **700**, and a blocking portion **356** that can cooperate with the groove **707** is provided on the convex piece **35**. The blocking portion **356** is provided obliquely. When the movable member **700** is rotated to the second position, the blocking portion **356** falls into the groove **707**, which can form a block for the movable member **700**, so that the movable member **700** can be accurately positioned in the second position.

(72) The preferred specific embodiments of the present invention have been described in detail

above. It should be understood that a person of ordinary skill in the art would be able to make various modifications and variations according to the concept of the present invention without involving any inventive effort. Therefore, any technical solution that can be obtained by a person skilled in the art by means of logical analysis, reasoning or limited trials on the basis of the prior art and according to the concept of the present invention should fall within the scope of protection defined by the claims.

Claims

1. A glue gun, comprising: a main body with one end formed as an accommodation part for accommodating a glue cylinder; a fixed handle connected to another end of the main body; a movable handle pivotally connected to the fixed handle; a push rod with one end located in the accommodation part; a brake provided on the push rod; and a movable member provided on the main body, wherein the movable member is configured to lock the brake, so as to continuously apply a force on glue in the glue cylinder to keep an internal stress of the glue, and release the brake to release the force applied on the glue in the glue cylinder, thereby releasing the internal stress of the glue; wherein a top of the main body is provided with a convex piece, and the movable member is pivotally connected to the convex piece; wherein the convex piece is provided with a pin shaft, a first through hole is provided in the movable member, and the pin shaft is sheathed in the first through hole.
2. The glue gun of claim 1, wherein the convex piece is provided with a limiting groove, the limiting groove has a first limiting end and a second limiting end, one end of the brake is located in the limiting groove, and the brake is configured to move between the first limiting end and the second limiting end.
3. The glue gun of claim 2, wherein the movable member is configured to be movable; and the movable member has a first position and a second position, wherein in the first position, the movable member locks the brake, so after the movable handle is released, the internal stress of the glue is still kept, so that the glue in the glue cylinder is dripping continuously; and in the second position, the movable member unlocks the brake, so after the movable handle is released, the internal stress of the glue is released, so that the glue does not drip any more.
4. The glue gun of claim 3, wherein the movable member has an end portion facing the brake; when the movable member is in the first position, the end portion locks the brake; and when the movable member is in the second position, the end portion releases the brake.
5. The glue gun of claim 3, wherein the first through hole is located in a middle of the movable member.
6. The glue gun of claim 5, wherein a second through hole is provided in the movable member, a positioning boss is provided on the convex piece, and when the movable member is in the first position, a part of the positioning boss is embedded into the second through hole.
7. The glue gun of claim 6, wherein the positioning boss is a trapezoid boss, side faces of the trapezoid boss are inclined, and a size of a top of the trapezoid boss is smaller than a size of a bottom of the trapezoid boss.
8. The glue gun of claim 5, wherein the movable member is provided with a corner portion, the corner portion faces the main body, and when the movable member is in the second position, the corner portion is in contact with the main body to block the movable member.
9. The glue gun of claim 8, wherein the movable member is provided with an arc-shaped portion, the arc-shaped portion and the corner portion are located on a same side of the movable member, and the arc-shaped portion is always in contact with the main body.
10. The glue gun of claim 5, wherein the movable member comprises a first part and a second part that are the same in shape and are symmetrically arranged, there is a gap between the first part and the second part, and the convex piece is inserted into the gap; and the first part and the second part

are connected together by a connection portion.
11. The glue gun of claim 10, wherein the connection portion is provided with a proximal end and a distal end, and the movable member is configured such that when the proximal end is pressed, the movable member is rotated to the first position, and when the distal end is pressed, the movable member is rotated to the second position.
12. The glue gun of claim 3, wherein a rotating shaft of the movable member is provided at an end away from the brake.
13. The glue gun of claim 12, wherein the movable member is configured such that when an end of the movable member opposite to the rotating shaft is pressed, the movable member is moved to the first position.
14. A glue gun, comprising: a main body with one end formed as an accommodation part for accommodating a glue cylinder; a fixed handle connected to another end of the main body: a movable handle pivotally connected to the fixed handle; a push rod with one end located in the accommodation part; a brake provided on the push rod; and a movable member provided on the main body, wherein the movable member is configured to lock the brake, so as to continuously apply a force on glue in the glue cylinder to keep an internal stress of the glue, and release the brake to release the force applied on the glue in the glue cylinder, thereby releasing the internal stress of the glue; wherein a top of the main body is provided with a convex piece, and the movable member is pivotally connected to the convex piece; wherein the movable member is provided on one side of the convex piece, and a rotating shaft of the movable member is provided at an end of the movable member away from the brake; wherein the rotating shaft of the movable member comprises a screw passing through the convex piece and a nut provided at one end of the screw, and the movable member is provided with a sleeve with which the nut is sheathed.
15. The glue gun of claim 14, wherein a blind hole is provided in the movable member, the blind hole and the sleeve are provided on a same side of the movable member, the convex piece is provided with a protrusion, and an elastic element connects the protrusion and the blind hole respectively.
16. The glue gun of claim 14, wherein the convex piece is provided with a limiting groove, the limiting groove has a first limiting end and a second limiting end, one end of the brake is located in the limiting groove, and the brake is configured to move between the first limiting end and the second limiting end.
17. The glue gun of claim 16, wherein the movable member is configured to be movable; and the movable member has a first position and a second position, wherein in the first position, the movable member locks the brake, so after the movable handle is released, the internal stress of the glue is still kept, so that the glue in the glue cylinder is dripping continuously; and in the second position, the movable member unlocks the brake, so after the movable handle is released, the internal stress of the glue is released, so that the glue does not drip any more.
18. The glue gun of claim 17, wherein a groove is provided in the movable member, a blocking portion is protruded on the convex piece, and the blocking portion falls into the groove when the movable member is in the second position.
19. The glue gun of claim 18, wherein the blocking portion is obliquely provided on the convex piece.
20. The glue gun of claim 17, wherein the movable member has an end portion facing the brake; when the movable member is in the first position, the end portion locks the brake; and when the movable member is in the second position, the end portion releases the brake.